

BioCrest - A Flexible Adaptive Robotic Hand Based on Idle-Stroke Mechanism *

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Abstract—Simultaneously achieving precision pinching and compliant grasping with active adaptation remains a significant challenge for most robotic hands and grippers. Furthermore, actively transitioning between grasping postures is difficult for hands or grippers with limited degrees of freedom (DOF). Even dexterous hands with high DOF often require extensive control strategies and extensive information from tactile sensors. This paper introduces BioCrest, a two-finger, three-DOF adaptive gripper exhibiting active reconfiguration capability. Each finger comprises the minimal number of components: three rigid links, one flexure strip, and one fingertip. Both fingers are actuated by a single shared motor. This simple yet unique finger structure enables both precise parallel pinching and highly compliant, stable grasping with uniform pressure distribution. A lost-motion transmission mechanism facilitates fingertip angle adjustment. Comprehensive kinematic and force analyses, combined with detailed mechanical design descriptions, clearly demonstrate controllable transition characteristics and stable grasping performance, including grasping force characteristics. Experiments validate the performance of the proposed gripper and demonstrate its practical utility in real-world applications.

I. INTRODUCTION

Robotic grasping and manipulation are fundamental capabilities required for robots to interact effectively with unstructured real-world environments. A key challenge in this domain lies in designing grippers that can seamlessly adapt to diverse object geometries and task requirements. Specifically, the simultaneous achievement of high-precision pinching and highly compliant, stable grasping with active adaptation represents a significant capability gap for most existing robotic hands and grippers[1], [2]. Precision pinching, crucial for handling small or fragile items, often demands rigid, parallel finger alignment and precise force control. Conversely, compliant grasping, essential for securing irregular, delicate, or variably shaped objects, necessitates passive or active conformation to object contours and distributed contact forces[3], [4], [5].

Many conventional rigid grippers excel at precise, parallel grasps but lack the inherent compliance needed for secure enveloping grasps[6]. While soft or underactuated grippers

offer excellent passive adaptability and stability for enveloping objects, they often sacrifice precise positioning and pinching capabilities[7], [8]. Moreover, actively transitioning between fundamentally different grasping postures such as switching from a parallel pinch to an enveloping grasp is particularly difficult, especially for grippers with limited degrees of freedom (DOF)[9]. This transition typically necessitates either complex multi-DOF mechanisms or specialized, often bulky, hardware.

Highly dexterous anthropomorphic hands possess the theoretical capability to perform both tasks and transition between them[10]. However, realizing this capability requires intricate, high-DOF mechanisms, extensive control strategies, and rich sensory feedback often including dense arrays of tactile sensors to manage complexity and ensure stable, robust manipulation[11], [12]. This complexity translates to high cost, control difficulty, and reduced reliability, limiting their widespread practical deployment[13], [14].

This paper introduces BioCrest, a novel two-finger, three-DOF adaptive robotic gripper designed to overcome these limitations through a remarkably simple yet highly effective mechanical architecture as shown in Figure. 1. The core innovation of BioCrest lies in its unique finger structure and actuation strategy: Each finger comprises only three rigid links, one flexure strip, and one fingertip. Crucially, both fingers are driven by a single shared motor. This minimalist design enables active reconfiguration capability, allowing the gripper to dynamically alter its grasping behavior *under a single actuator*. The flexure strip provides essential structural compliance, while a carefully designed lost-motion transmission mechanism facilitates independent fingertip angle adjustment during actuation.

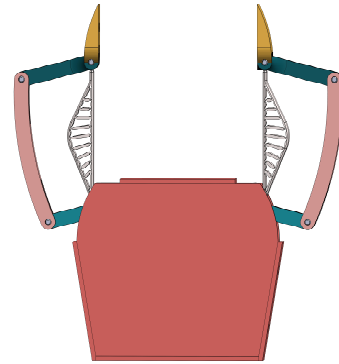


Figure 1. The BioCrest gripper.

As a direct result of this novel mechanical design, BioCrest demonstrates:

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1. High-precision parallel pinching with rigid finger alignment.
2. Highly compliant and stable enveloping grasping with uniform pressure distribution across the contact surfaces.
3. Active and controllable transition between these distinct grasping modes.

Section II details the implementation of the grasping modes. Section III describes the design of the BioCrest finger mechanism. Section IV provides a comprehensive kinematic and mechanical analysis outlining the supporting gripper functions and stability, elucidating – through the complete mechanical design – the principles enabling its controllable switching capability and stable grasping performance, including grasping force characteristics. Subsequently, Section V reports the conducted grasping experiments. Finally, Section VI summarizes the entire paper.

II. IMPLEMENTATION OF THE GRASPING MODES

A. Precision Parallel Pinch Mode

The BioCrest adaptive two-finger gripper features two switchable grasping modes: Precision Parallel Pinch and Self-adaptive Enveloping Grasp.

In Precision Parallel Pinch mode, the gripper employs a topologically optimized parallelogram linkage mechanism (illustrated in Fig. 2) to maintain strictly parallel alignment of both fingers throughout the opening/closing stroke. This enables millimeter-precision grasping of thin-walled or regular-shaped objects such as PCB boards, bolts, and chip trays. Crucially, the base link of this parallelogram mechanism is not rigidly fixed but flexibly connected to the base via a pair of preload-adjustable stainless steel torsion springs. When subjected to lateral disturbances or transient impacts, this spring-damper system absorbs and dissipates energy. This design ensures virtual fixation of the parallelogram base while significantly reducing mechanism backlash and wear, maintaining exceptional repeatability in the arc trajectory. The spring preload force is continuously adjustable within a 2–5 N range via the base rear during assembly to accommodate varying payload requirements, from lightweight foam to metal sheets.

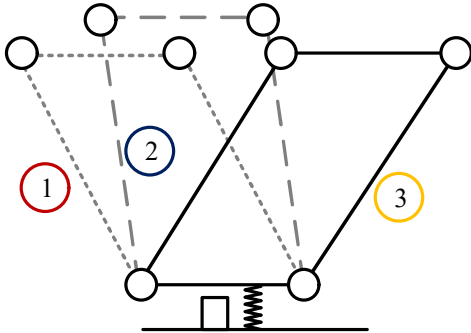


Figure 2. Parallelogram Linkage Mechanism.

B. Self-adaptive Enveloping Grasp Mode

For target objects exhibiting variable geometric dimensions and complex surface curvatures—such as fruits, toys, or irregular castings—the BioCrest gripper seamlessly

transitions into Self-adaptive Enveloping Grasp mode. In this mode, the drive motor actively reduces output stiffness, enabling the finger assembly comprising rigid base links and the flexible phalanx segments to fully exploit passive compliance. Concurrently, the parallelogram linkage serving as the gross opening/closing actuator assumes a secondary role: its function shifts to providing initial grasp configuration adjustment rather than forcibly constraining fingertip orientation degrees of freedom.

A core innovation lies in its integrated Idle-Stroke mechanism. This mechanism consists of two symmetrically arranged gears with specific-angle protrusions, engaged with a link featuring a mating protrusion as shown in Figure 3. Its fundamental principle is an initial, precise, and symmetrically designed backlash angle at the cam interface.

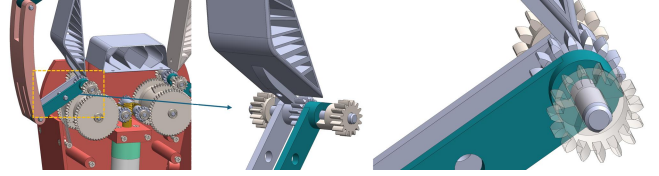


Figure 3. The Idle-Stroke mechanism.

In Precision Parallel Pinch mode, input torque first consumes the backlash between gears while protrusions remain non-contacting, maintaining rigid mechanism behavior. During transition to Adaptive Enveloping Grasp, however, input torque drives the cam link—i.e., the base link of the parallelogram linkage—compelling protrusion contact and interaction. This actively induces bending deformation at the fingertip joint, achieving adaptive envelope and force closure around the target object as shown in Figure 4.

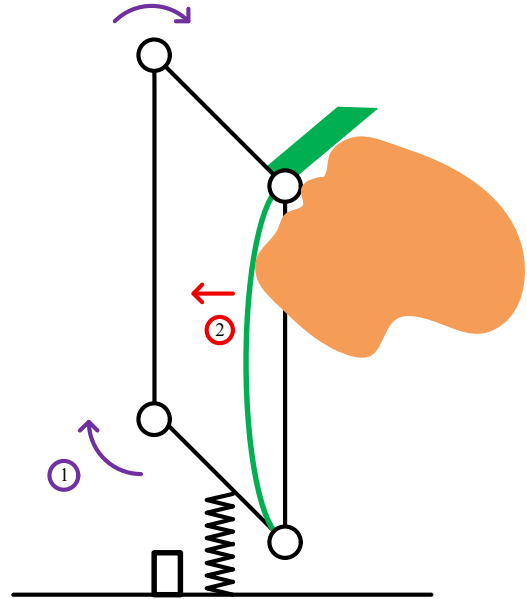


Figure 4. The Idle-Stroke mechanism.

This intelligent bimodal switching mechanism (rigid parallel pinching vs. compliant enveloping) autonomously adjusts based purely on the mechanism's inherent kinematic and dynamic properties, requiring neither complex control algorithms nor auxiliary actuators. Consequently, BioCrest

successfully integrates manufacturing-demanded high-precision positioning/assembly capabilities with logistics/warehousing-required strong robust adaptive handling performance within a single end-effector unit.

III. DESIGN OF THE BIOCREST FINGER

A. Design of the flexure strip

The Biocrest adaptive gripper comprises a pair of mirror-symmetrical fingers. Each finger is structurally divided into three rigid links and one flexible phalanx. All structural components are fabricated using fused deposition modeling (FDM), ensuring lightweight design requirements while significantly reducing costs; each link is interconnected via low-friction bearing hinges, ensuring smooth motion and minimizing return clearance. The flexible phalanx is located at the connection between the fingertip and the base, corresponding to the compliant flexible structure depicted in Fig. 5. Its internal honeycomb topology enables controlled inward concave deformation upon compression by a target object. The internal hollow structural elements within the flexible segment measure only 0.8 mm in thickness, facilitating enhanced bending performance., thereby passively adapting to envelope objects of varying shapes, sizes, and stiffness.



Figure 5. Design of the flexible phalanx and its fabricated sample..

This flexible segment utilizes thermoplastic polyurethane (TPU) with a hardness of 95 A and is manufactured in a single step via multi-nozzle FDM printing, with the printing path topologically optimized to maintain compliance while enhancing tear resistance and elastic recovery lifespan. Due to its rigid-flexible coupled design philosophy, Biocrest achieves robust, high-tolerance, low-damage grasping without complex force-control algorithms, making it suitable for diverse applications including fruit and vegetable sorting, irregular workpiece handling, and fragile item transportation.

B. Structural Configuration

The Biocrest finger mechanism integrates two core components: the flexible phalanx (FP) and the lost-motion transmission mechanism, as depicted in Fig. 6. This mechanical system comprises three primary elements: the Distal Phalanx (DP), the Flexible Phalanx (FP), and the Cam-Driven Linkage (CDL). DP and FP are kinematically coupled through a distal joint shaft, enabling the FP to execute planar translation relative to the fixed base during parallel

pinching while retaining autonomous rotational capability. During static operation, tension spring S kinematically couples Shaft 1 and the base to maintain spatial congruence between the proximal and medial joint axes. This configuration enables the Biocrest finger to deliver stable parallel pinching.

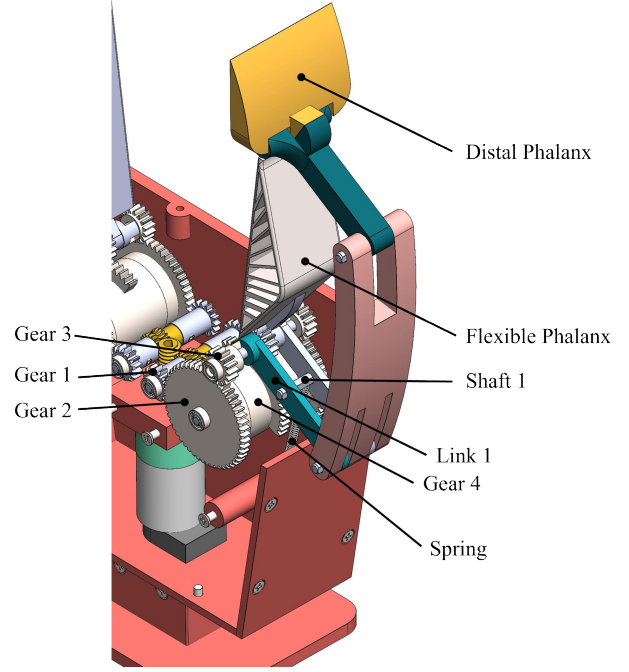


Figure 6. The Design of the Biocrest finger.

Notably, the torsion spring integrated within Gear 2 features a left-handed coil structure, while its counterpart within the opposing gear incorporates a right-handed coil torsion spring. These two torsion springs possess identical specifications but opposite coil orientations, as depicted in Fig. 7. Collectively, they constitute the core functionality of the lost-motion transmission mechanism: When the Biocrest finger executes an inward motion, the left-handed torsion spring in Gear 2 and Gear 4 immediately generates corresponding left-handed torque, while the right-handed torsion spring in the opposing gear simultaneously produces counteracting right-handed torque. These equal-magnitude, opposite-direction torques mutually counteract, eliminating backlash-induced fluctuations and ensuring the fingertip output force remains consistently smooth and constant—devoid of any sudden force fluctuations or jerky motion.

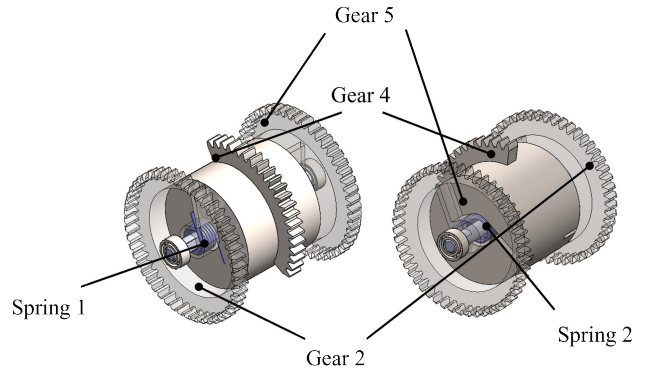


Figure 7. The Idle-Stroke Gear.

C. Operational Modes

As experimentally validated in Fig. 6, the Biocrest gripper exhibits dual-mode grasping functionality that is mechanically deterministic and entirely governed by the intrinsic kinematics of its transmission architecture:

a) Parallel grasping mode (Fig. 6a). When the task involves relatively regular elongated objects or small, standardised items, the primary transmission system enforces a rigid, closed kinematic loop that constrains the distal phalanges to remain strictly parallel throughout the closing trajectory. This geometric constraint is delivered by the coupled worm-gear pair and the four-bar linkage formed by the actuation rod, ensuring that the line of action of each fingertip stays invariant relative to the palm reference frame.

b) Active-adaptive grasping mode (Fig. 6b). Upon encountering large or irregularly shaped targets, the secondary transmission path is engaged once the initial parallel motion is obstructed. The stored 90° idle stroke is consumed, and the CDL mechanism actuates the secondary phalanges to rotate inward in a controlled, torque-dependent manner. This rotation introduces passive compliance without requiring sensory feedback, enabling the phalanges to conform to the local surface curvature of the object and thereby maximise contact area and grasp stability.

These two operational regimes are mutually exclusive yet seamlessly switchable within a single grasping cycle, providing the gripper with the capacity to accommodate a broad spectrum of object geometries through purely mechanical means.

The finger mechanism demonstrates two distinct operational modes, as shown in Fig. 7. The underlying kinematic analysis reveals:

The BIOCREST finger integrates dual transmission systems. The primary system, composed of a worm-gear pair, active gear, passive gear, and actuation rod, executes parallel grasping. Motor torque is delivered through the worm-gear pair to the active gear, which meshes with the passive gear to drive the actuation rod. The rod's arc motion is translated into the linear motion of the secondary phalanx.

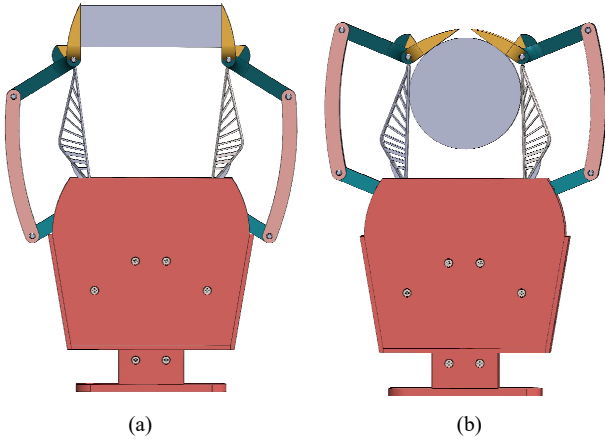


Figure 8. Two grasping modes of the Biocrest gripper. (a) Linear parallel pinching for precision grasp; (b) Self-adaptive grasping for power grasp.

The secondary transmission system, consisting of a worm-worm-gear pair, driving gear, driven gear, component C1, and a CDL mechanism, enables active, shape-adaptive grasping. Motor torque is transmitted through the worm pair to sequentially drive the driving gear, the driven gear, and finally C1. A 90° idle stroke is maintained between C1 and the CDL mechanism. During parallel pinching, the CDL mechanism idles without cam-block engagement. Upon finger convergence or object contact, rotation of cam wheel 1 brings the cam blocks into engagement, actuating the CDL mechanism. This actuation deforms the parallelogram linkage—wherein the secondary phalanx is attached to the upper bar of the kinematic chain—inducing active inward bending of the secondary phalanx through the transmission, thereby achieving precise shape conformity.

IV. KINEMATIC ANALYSIS OF THE BIOCREST FINGER

This section details the kinematic analysis for the proposed robotic finger mechanism. Employing the Complex Vector Method, the finger's coordinate system is rigorously established to determine precise linkage angles during actuation and thereby characterize its feasible workspace.

A. Kinematic Analysis

The Biocrest system incorporates a planar parallelogram mechanism as illustrated in Fig. 9. This mechanism comprises four links: AB, BC, CD, and DA. Links AB and CD are of equal length, and the base AD is fixed and also of identical length. Points A and D are fixed to the ground. The mechanism is actuated by driving link AB with constant angular velocity about point A.

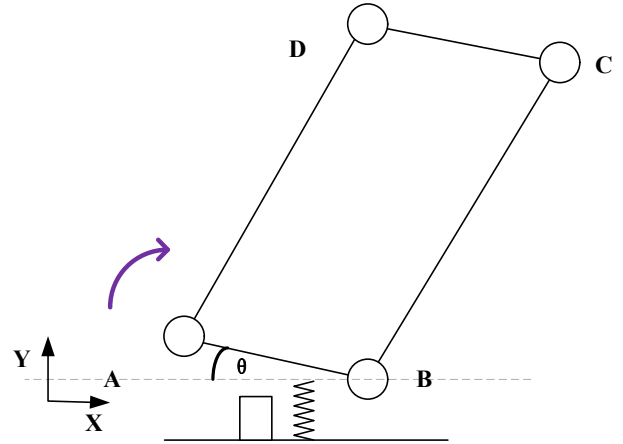


Figure 9. The planar parallelogram mechanism.

The objective is to analyze the motion of point C (the upper-right corner) over the range of motion where the angle θ between link AB and the positive x-axis varies from $\theta_{\min}$ to $\theta_{\max}$. A fixed coordinate system is defined with point A at the origin and point D located at $[X_D, 0]$.

The position of point C, the endpoint of link CD, can be described as follows: Since CD is a rigid body of fixed length and point D is fixed at $[X_D, 0]$, point C must lie at a distance equal to the length of CD from point D. Furthermore, due to the parallelogram configuration, link CD maintains the same orientation as link AB. Therefore, the angle that CD makes

with the horizontal axis is also θ . Consequently, the coordinates of point C are:

$$x_c = L_2 + L_1 \cos \theta \quad (1)$$

$$y_c = L_1 \cos \theta \quad (2)$$

Where $\theta = \omega t$ and t is time

The path traced by point C is obtained by eliminating the parameter θ (or time t):

$$(x_c - L_2)^2 + y_c^2 = L_1^2 \quad (3)$$

Differentiating position equations with respect to time yields velocity components:

$$v_x(t) = \dot{x}_c = -L_1 \omega \sin \theta(t) \quad (4)$$

$$v_y(t) = \dot{y}_c = L_1 \omega \cos \theta(t) \quad (5)$$

The resultant speed is calculated as:

$$v(t) = \sqrt{v_x^2(t) + v_y^2(t)} = L_1 \omega \quad (6)$$

The velocity vector is always perpendicular to vector DC:

$$\vec{v}_C \perp \vec{DC} = \begin{bmatrix} x_c - L_2 \\ y_c \end{bmatrix} \quad (7)$$

The direction angle ϕ relative to the positive x-axis is:

$$\phi(t) = \tan^{-1} \left(\frac{v_y(t)}{v_x(t)} \right) + \pi = \tan^{-1} \left(\frac{\cos(t)}{-\sin(t)} \right) + \pi \quad (8)$$

Based on the iterative computational process outlined in Equations (1)-(8), Fig. 10 demonstrates the motion trajectory and the characteristics of the Biocrest finger and the table.

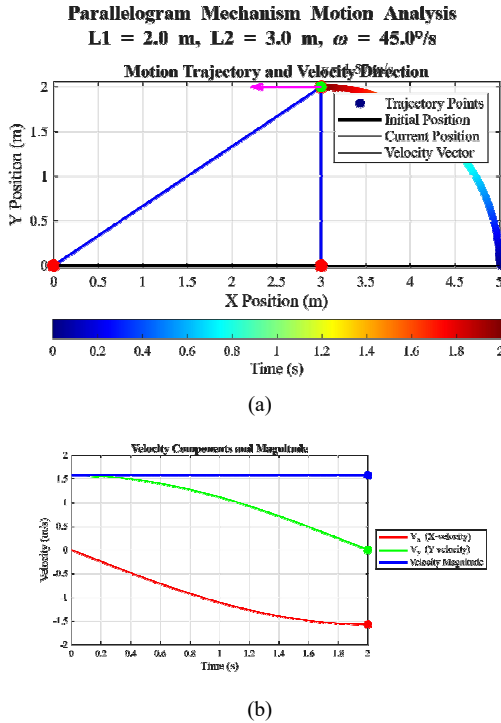


Figure 10. Analysis of trajectory and velocity of C : (a) Parallelogram Mechanism Motion Analysis, (b) Velocity Components and Magnitude.

TABLE I. PHYSICAL SYMBOLS AND THEIR MEANINGS

Symbol	Description	Units
A	Fixed origin point	m
D	Fixed base pivot	m
$L1$	AB and CD link lengths	m
$L2$	Base AD length	m
θ	Angular position of AB	rad
ω	Angular velocity of AB	rad/s
x_C, y_C	Coordinates of point C	m
v_x, v_y	Velocity components of C	m/s
v	Speed of point C	m/s
ϕ	Velocity direction angle	rad

V. GRASPING EXPERIMENTS OF THE BIOCREST GRIPPER

To verify design feasibility, Biocrest was manufactured using FDM and features a compliant finger segment that emulates the biomechanics of the human palm while maintaining 5mm deformation compliance for irregular object accommodation.

The parallel pinch mode employs a parallelogram mechanism, preserving gripper orientation during linear displacement. This operational characteristic is critical for high-precision applications such as industrial pick-and-place operations or delicate object manipulation in medical robotics.

Biocrest's gripper utilizes dual independently actuated motors to drive symmetrically arranged fingers, providing dimensional adaptability and balanced force distribution. Its time-delayed active adaptation mechanism enables stable shape-adaptive enveloping grasps, targeting industrial automation and domestic service robotics.

As shown in Fig. 11, the gripper successfully handled a pen, tape roll, and compact speaker via dual-mode operation: parallel grasping maintains linear fingertip alignment for symmetrical targets, while adaptive grasping achieves geometric self-adaptation for irregular shape stabilization..

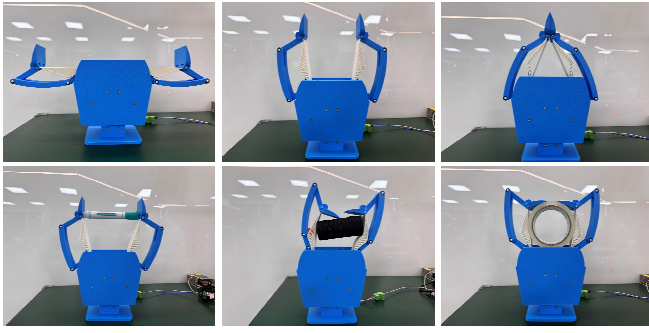


Figure 11. Experiments of Biocrest gripper.

The adaptive grasping of the Biocrest gripper was experimentally validated using specified test objects. Results demonstrate the secondary phalange adaptively conforms to object profiles with simultaneous primary phalange tilting, confirming adaptive mechanics. This behavior stems from passive compliance through delayed active adaptation and precise linkage control, enabling geometric reconfiguration.

VI. CONCLUSIONS

This paper has presented BioCrest, a novel two-finger adaptive gripper that overcomes the longstanding challenge of integrating high-precision parallel pinching and compliant enveloping grasping within a mechanically deterministic, single-actuator architecture. Through its idle-stroke transmission mechanism and minimalist structural design—comprising only three rigid links, one flexure strip, and a shared motor per finger—BioCrest achieves active reconfiguration between rigid parallel alignment and passive compliance modes without requiring complex control algorithms or dense sensor arrays. Experimental validation confirms the gripper's dual-mode capability: the parallelogram linkage ensures millimeter-precision linear trajectory for delicate objects (e.g., PCB boards), while the flexure strip's honeycomb topology enables exceptional shape adaptation and uniform pressure distribution during enveloping grasps of irregular items (e.g., fruits or castings). Crucially, the transition between these modes is autonomously governed by intrinsic kinematic properties, as demonstrated in grasping tests spanning standardized and geometrically diverse targets.

These results substantiate BioCrest's potential to bridge industrial precision requirements and logistics-oriented robustness demands in a cost-effective, reliability-optimized end-effector solution.

REFERENCES

- [1] F. Liu, F. Sun, B. Fang, X. Li, S. Sun, and H. Liu, "Hybrid Robotic Grasping With a Soft Multimodal Gripper and a Deep Multistage Learning Scheme," *IEEE Trans. Robot.*, vol. 39, no. 3, pp. 2379–2399, June 2023.
- [2] B.-Y. Sun *et al.*, "Design Principle of a Dual-Actuated Robotic Hand With Anthropomorphic Self-Adaptive Grasping and Dexterous Manipulation Abilities," *IEEE Trans. Robot.*, vol. 38, no. 4, pp. 2322–2340, Aug. 2022.
- [3] J. Zhou, X. Chen, J. Li, Y. Tian, and Z. Wang, "A soft robotic approach to robust and dexterous grasping," in *2018 IEEE International Conference on Soft Robotics (RoboSoft)*, Apr. 2018, pp. 412–417.
- [4] G.-C. Jeong, A. Bahety, G. Pedraza, A. D. Deshpande, and R. Martín-Martín, "BaRiFlex: A Robotic Gripper with Versatility and Collision Robustness for Robot Learning," in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Oct. 2024, pp. 4106–4113.
- [5] V. Babin and C. Gosselin, "Mechanisms for Robotic Grasping and Manipulation," May 2021.
- [6] X. Zhang and Q. Xu, "Design and analysis of a constant-force parallel micro-gripper," in *2017 IEEE International Conference on Robotics and Biomimetics (ROBIO)*, Dec. 2017, pp. 1112–1117.
- [7] C.-H. Liu, Y.-C. Shih, and J.-C. Chi, "Topology Optimization and Prototype of a Highly Adaptive Multi-Material Compliant Finger for Parallel Grippers," *IEEE Robot. Autom. Lett.*, vol. 9, no. 9, pp. 8067–8074, Sept. 2024.
- [8] X. Li *et al.*, "Porous Magnetic Soft Grippers for Fast and Gentle Grasping of Delicate Living Objects," *Adv. Mater.*, vol. 36, no. 44, p. 2409173, 2024.
- [9] L. Li *et al.*, "A Comparative Analysis and Scoping Review of Soft–Rigid and Industrial Parallel Rigid Grippers," *Adv. Intell. Syst.*, vol. 7, no. 5, p. 2400503, 2025.
- [10] T. Nishimura, T. Muryoe, and T. Watanabe, "High-Payload and Self-Adaptive Robotic Hand With 1-Degree-of-Freedom Translation/Rotation Switching Mechanism," *IEEE Robot. Autom. Lett.*, vol. 8, no. 5, pp. 2470–2477, May 2023.
- [11] N. Ribeiro *et al.*, "RAMBO — Robotic Anthropomorphic Manipulator for Bimanual Operations," in *2024 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*, May 2024, pp. 157–164.
- [12] M. K. Gonzalez, N. A. Theissen, A. Barrios, and A. Archenti, "Online compliance error compensation system for industrial manipulators in contact applications," *Robot. Comput.-Integr. Manuf.*, vol. 76, p. 102305, Aug. 2022.
- [13] E. Abbot, A. de Oliveira Barros, and J. Yang, "Novel Design of a 3D Printed Anthropomorphic Soft Prosthetic Hand,".
- [14] C. Zhang *et al.*, "An Anthropomorphic Robotic Hand With a Soft-Rigid Hybrid Structure and Positive-Negative Pneumatic Actuation," *IEEE Robot. Autom. Lett.*, vol. 8, no. 7, pp. 4346–4353, July 2023.